Marshall, Sebastian Rushawn Odaine

Coastal & Hydroinformatics Engineer | Hazard-Resilient Infrastructure & AI/ML-driven Hydrology

Ph.D. candidate in Civil Engineering – University of Virginia, Charlottesville, 22904 VA, USA

Email: xec3cr@virginia.edu | **Languages**: English, Mandarin Chinese | Google Scholar | LinkedIn

PROFESSIONAL SUMMARY

Licensed Coastal & Hydroinformatics Engineer (P.E.) and current Ph.D. researcher at the University of Virginia with a proven track record of leading climate-resilience projects (up to US $3 million value), delivering shoreline-protection and flood-hazard models that safeguard the welfare of substantial populations (over 250,000 Caribbean residents) and protect critical infrastructure. Authored three peer-reviewed papers (80+ citations) on AI/ML-enabled hydro-climate forecasting and compound-flood risk, demonstrating expertise in cutting-edge, transferable methodologies. My goal is to apply advanced data-assimilative modeling, remote sensing, and policy-oriented engineering to advance U.S. coastal and riverine infrastructure in the national interest, protecting American lives, property, and critical economic assets from accelerating climate hazards.

EDUCATION & TRAINING

2025 - present **Ph.D. in Civil Engineering - Environmental & Water Resources Engineering** – University of Virginia, VA, USA

2020 - 2022 **MSc. in Hydroinformatics & Water Management** – (EuroAquae+) Europe

Erasmus Mundus (European Union Scholarship) Joint Master's Degree Program | GPA: 3.7/4.0

- Université Côte d'Azur, Nice, France (08/2020 - 01/2021)
- Newcastle University, Newcastle, the United Kingdom (02/2021 - 08/2021)
- Brandenburg University of Technology, Cottbus-Senftenberg, Germany, (09/2021 - 09/2022)

Courses: Hydraulics & Hydrology, Flood Modeling & Forecasting, GIS-based tools, Groundwater Modeling, Computational Hydraulics, Integrated River Basin Management.

2015 - 2020 **B.E. in Harbor, Waterway, and Coastal Engineering** – Hohai University

Hohai University | GPA: 4.7/5.0 | Nanjing, China.

Courses: Soil and Structural Mechanics, River Hydrodynamics, Hydraulic and Water Conservation, Port Hydraulic Structures, Surveying, Estuarine and Coastal Evolution, Construction Project Management.

PROFESSIONAL SOCIETY MEMBERSHIPS

- American Society of Civil Engineers (ASCE)
- American Water Works Association (AWWA)
- American Geophysical Union (AGU)
- Environmental and Water Resources Institute (EWRI)
- Coasts, Oceans, Ports & Rivers Institute (COPRI)
- Jamaica Institution of Engineers (JIE)

PROFESSIONAL EXPERIENCE

11/2022–Present
Kingston, Jamaica

Smith Warner International Ltd.

Junior Coastal and Hydroinformatics Engineer

- As a Junior Coastal Engineer at Smith Warner International, I design coastal structures and conduct numerical modeling to support climate adaptation and hydrological projects. My expertise spans a variety of tasks, including the development of shoreline protection solutions, utilizing MIKE 21 and HEC-RAS to model wave dynamics, flood risk, and sediment transport. I also lead hydrological data analysis and hazard mapping for vulnerable watersheds, contributing to projects designed to mitigate the impacts of storm surges and rising sea levels. I work closely with multidisciplinary teams, ensuring the integration of remote sensing data with GIS and Civil 3D for design drafting, while also preparing detailed bills of quantities and technical specifications for project implementation. Furthermore, I oversee construction supervision, ensuring project compliance with engineering standards and environmental regulations. My contributions have been instrumental in delivering several major flood risk management and coastal defense projects, with successful completion of projects valued between $500,000 to $3 million USD.

07/2021–10/2021
Remote, France

Data/Geospatial Analyst (Lead)

- As the Lead Data/Geospatial Analyst, I directed a team of five interns across two countries, managing all phases of geospatial data analysis while working remotely from France. Using QGIS, I spearheaded the analysis of large-scale datasets to identify geographic patterns critical for environmental planning. I was responsible for developing data collection systems and optimizing analytics strategies, which resulted in a 75% improvement in data accuracy and a 50% reduction in operational delays. My collaboration with senior management enabled me to align business requirements with data-driven solutions, resulting in improved process efficiency. Additionally, I supervised interns, ensuring adherence to timelines and the effective implementation of data-related strategies.

07/2017–08/2017
Kingston, Jamaica

Intern Coastal / Environmental Engineer

- The core of my internship involved the use of a Hydraulic Modelling tool, Delft 3D, to construct 3D Flexible meshes within the South American region based on bathymetric and surveying data to investigate hydrodynamics, sediment transport, and morphology. The project explored alternative software to reduce reliance on a single tool, improving project efficiency by 20% and reducing wait times by 15%.

03/2022–08/2022

SUEZ - Smart & Environmental Solutions

Intern Hydraulic / Smart Water Engineer

Paris Ile-de-France, France.

- Implemented a Virtual Sectorisation model on the AQUADVANCED® Water Networks platform for a $1M project in Milan, Italy, and Dijon, France. Conducted water network diagnostics for leak detection and operational anomalies. Modeled hydraulics and hydrology while developing a digital platform and implementing GIS systems. Managed databases using C#, Python, MATLAB, and SQL, and drafted client documentation. Improved water network management efficiency by 35%, reducing operational anomalies and improving response times.

PROFESSIONAL SKILLS/ LICENSE

License

- Registered Professional Engineer (P.E.) | Reg#: PE/01,14/01399 | Discipline: Civil & Marine

Areas of Expertise

- Remote Sensing, AI/ML in Hydrology, Climate-Resilient Infrastructure Design, Beach Creation and Coastal Stabilization, Numerical Modeling of coastal processes (wave dynamics, flood risk, sediment transport using DHI suite, HEC-RAS, etc.), Design of Marine & Coastal Structures (piers, jetties, breakwaters), Hydrological Modeling (SWAT, HEC-HMS), Coastal Zone Management and Planning, Hazard Analysis and Risk Assessment, Climate Change Capacity Building, Policy-Oriented Engineering.

Organizational/ Personal skills

- Leadership, decision-making, coordinating, planning, communication, critical thinking, detailed-oriented, multi-tasking, problem-solving, time management skills, enthusiastic, energetic, curious, able to deal with: uncertainty, changes in scope, and ambiguity.

Computer & Job-related skills

- HEC-RAS, HEC-HMS, SWAT, ENERCALC, Hydraulic Modelling Software's (DHI suite, HEC-RAS, etc.), AutoCAD, Programming Software's: C++, Java, MATLAB, Python, R Programming, QGIS, Microsoft Word, Excel, Access, PowerPoint, Outlook., Project Management: Microsoft Project, Total Station, Optical Theodolite, Optical Level.

Foreign language

Mandarin Chinese	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
	C1	B2	C1	C1	B1
	The Hanyu Shuiping Kaoshi / Chinese Proficiency Test (HSK) 4 / 6				

SCHOLARSHIPS/ FELLOWSHIPS

Years	Award	Stipend Value	Competitiveness & Scope
2025 – 2030	Graduate Researcher - Ph.D. Civil & Environmental Engineering, Univ. of Virginia	$66,000 per year (stipend + health-insurance subsidy)	Selected one of fewer than twelve scholars worldwide issued annually; supports cutting-edge research in Environmental & Water Resources Engineering while waiving all tuition and mandatory fees.
2020 – 2022	Erasmus Mundus Joint Master's Scholarship - Hydroinformatics & Water Management (EuroAquae+)	€49,000 total stipend	Selected 1 of 12 scholars worldwide by the European Union; stipend complemented by a complete tuition waiver and inter-EU mobility funding.
2014 – 2019	China Harbour Engineering Company (CHEC) Scholarship - B.E. Harbor, Coastal & Waterway Engineering	$73,400 total stipend	Selected 1 of 15 scholars worldwide; stipend accompanied by full tuition, accommodation, and insurance coverage provided jointly by CHEC (Americas Division) and China Communications Construction Company (CCCC).

AWARDS & HONORS

2019 Outstanding Graduation Design of Shengsi Maguan Stagnant Current and Wave Breakwater Design Project. Awarded by Hohai University, China.

2018 Chinese Government's Haiyun Fenghua Outstanding Student Award. Awarded by Hohai University, China.

2017 China Harbour Engineering Company Outstanding Academic Excellence Award. Awarded by Hohai University, China.

2017 Outstanding Academic International Student Award. Awarded by Hohai University, China.

2015 Outstanding Academic Excellence Honorary Award. Awarded by Nanjing Normal University, China.

PROJECTS

2022 **Master's Thesis: Implementation of a Virtual Sectorization System for Real-Time Drinking Water Network Management**

Developed a standardized procedure for deploying AQUADVANCED® Water Networks software in urban water systems (case studies in Dijon, France and Milan, Italy). The project involved integrating a hydraulic model, configuring the platform for real-time anomaly detection, and drafting technical guidelines for large-scale implementation. This work demonstrated how virtual sectorization and continuous performance monitoring can reduce water losses, optimize resource usage, and enhance overall water distribution efficiency.

2019 **Bachelor's Thesis: Shengsi Maguan Stagnant Current and Wave Breakwater Design Project**

The Shengsi Maguan Detention and Wave Protection Project in Zhejiang Province is located on the SE side of Sijiao Island in Shengsi County, Zhejiang Province. The elevation of the Bay bottom is generally about - 2m to - 4m. Due to the whole bay environment being conducive to Aquaculture to develop the Aquaculture industry further, a Stagnation and Wave Protection project is proposed to be built to reduce the impact of wind waves mainly from the SE direction and ultimately form a new and relatively calm Aquaculture Environment. The main contents of this paper include Hydrological Analysis and Structural Design Calculation.

SERVICES & PROFESSIONAL ACTIVITES

Year	Description	Organization / Journal
2025 - current	**Academic Manuscript Reviewer**	Earth Systems and Environment *(Springer Nature)*, Journal of Hydrology: Regional Studies, Discover Sustainability *(Springer Nature)*

PUBLICATIONS [*corresponding]

2025 **Sebastian R.O. Marshall**, Thanh-Nhan-Duc Tran*, Mahesh R. Tapas, Binh Quang Nguyen. Integrating Artificial Intelligence and Machine Learning in Hydrological Modeling for Sustainable Resource Management. *International Journal of River Basin Management*. *https://doi.org/10.1080/15715124.2025.2478280*

Sebastian R.O. Marshall, Thanh-Nhan-Duc Tran* and Venkataraman Lakshmi. SWAT and CMIP6-driven hydro-climate modeling of Assessing future flood risks and vegetation dynamics in the White Oak Bayou Watershed, United States. *Earth System and Environment*. *Remote Sensing. (2023)*, *https://doi.org/10.1007/s41748-025-00621-2*

2022 Tran T-N-D, Nguyen QB, Vo ND, **Marshall R***, Gourbesville P. Assessment of Terrain Scenario Impacts on Hydrological Simulation with SWAT Model. Application to Lai Giang Catchment, Vietnam. *Advances in Hydroinformatics. Springer Water (2022):1205-1222. https://doi.org/10.1007/978-981-19-1600-7_77*

PUBLICATIONS UNDER REVIEW & PREPARATION

2025 Amgoth Ashok, **Sebastian R.O. Marshall*,** Mahesh R. Tapas1, Thanh-Nhan-Duc Tran, Md Masudur Rahman. Comparative Analysis of Machine Learning Classifiers for Land Use and Land Cover Mapping: A Case Study of Warangal District, India. *Journal of the Indian Society of Remote Sensing* (***under review***)

Sebastian R.O. Marshall, Thanh-Nhan-Duc Tran*, Binh Quang Nguyen, Arfan Arshad, Venkataraman Lakshmi. Decadal Trends in Meteorological Variables and Streamflow under Climate Scenarios. *Journal of Hydrology: Regional Studies. (**major revision - resubmitted**)*

Sebastian R.O. Marshall, Thanh-Nhan-Duc Tran*, Venkataraman Lakshmi. NASA Land Information System (LIS) Systematic Review: Advancing Land Surface Modeling and Data Assimilation for Hydrology and Earth Observations. *Journal of Hydrology: Regional Studies. (**major revision - resubmitted**)*

2024 Tapas R. M, **Sebastian R.O. Marshall**, Tran T-N-D* and Lakshmi V. Chap. 27. The Climate and Land-Use Change Feedback for Congo Basin, Central Africa. *Science Panel for the Congo Basin (SPCB)*. (***under review***)

CONFERENCES [*presenter]

2025 **Sebastian R.O. Marshall***, Thanh-Nhan-Duc Tran, ManhHung Le, Binh Quang Nguyen, Bin Fang, Hesham Elhaddad, John B Eylander, Sameh Kantosh, Sydney Crisanti, John D Bolten and Venkataraman Lakshmi. Global Assessment of Compound Meteorological-Soil Moisture Droughts: Trends and Climate Mode Influences, United States. *AGU '2025.*

Sebastian R.O. Marshall*, Thanh-Nhan-Duc Tran and Venkataraman Lakshmi. Operational Hydrological Modeling in Data-Scarce Regions: Assessing the Impact of Fine-Resolution Soil Moisture Assimilation. *AGU '2025.*

2024 **Sebastian R.O. Marshall***, Thanh-Nhan-Duc Tran and Venkataraman Lakshmi. SWAT and CMIP6-driven hydro-climate modeling of Assessing future flood risks and vegetation dynamics in the White Oak Bayou Watershed, United States. *AGU '2024.*

REFERENCES

Dr. Venkataraman (Venkat) Lakshmi
Venkataraman Lakshmi | Google Scholar
John L. Newcomb Professor of Engineering, University of Virginia, 22904 VA, USA, Head of Global Hydrology and Water Resources Group.
Email: vl9tn@virginia.edu

Dr. Michael Cosh
Michael Cosh | Google Scholar
Research Hydrologist - Research Leader, USDA-ARS, Hydrology and Remote Sensing Laboratory, Beltsville, MD
Email: Michael.Cosh@usda.gov

Dr. John D. Bolten
John D. Bolten | Google Scholar
Chief of the Hydrological Sciences Laboratory, NASA Goddard Space Flight Center (GSFC), MD 20771, USA
Email: john.bolten@nasa.gov